

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS:

Total Marks: 20

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I M Meghanadh(192325026)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Meghanadh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

CO2-AT3 - Class Test

1) Introduction: Machine learning (ML) and Deep learning are two important branches of artificial intelligence (AI). Both enable computers to learn from data and make predictions or decisions without being explicitly programmed.

Deep learning is a specialized subset of machine learning that uses artificial neural networks.

with multiple layers to solve complex problems.

Difference b/w machine learning (ML) and Deep learning

Aspect	Machine learning (ML)	Deep learning (DL)
Architecture	uses simple algorithms such as Decision Trees, SVM, Naive Bayes, and Random Forest.	uses multi-layered (ANNs) especially Deep neural networks (DNNs).
Data Requirements	works well with small to medium-sized datasets.	Requires very large datasets for high accuracy.
Feature extraction	Features are manually selected and engineered by humans.	Features are automatically learned from raw data by the neural networks.
Applications	Suitable for structured data like tabular datasets.	Best for unstructured data such as images, audio and text.

Real-world Applications:

Machine Learning

- 1) Email Spam Detection - Classifies emails as spam or not
- 2) Credit Card Fraud Detection - Detects fraudulent financial transaction

Deep Learning:

- 1) Face Recognition System - Used in smart phone face unlock and security systems.
- 2) Autonomous Vehicles - Detects roads, pedestrians, traffic signs.

Summary:

- Machine learning relies on manual feature engineering and performs well with smaller datasets.

2A) Introduction

Representation Learning is a technique in Deep Learning where a neural network automatically learns useful features from raw data instead of relying on manually designed features. It enables the model to identify important patterns and representations that improve prediction accuracy. This approach is widely used in image recognition, speech.

Advantages:

- Learns features automatically
- Reduce manual effort.

width of a Neural Network

Influence:

- Learning capability: Learn more features at the same level.
- Computational complexity: Requires more memory and computation.

Depth of Neural Network:

Influence:

- Learning Capability: Learns simple to complex feature hierarchically.
- Computational complexity: Increases training time and computational cost.

Comparison:

Aspect	Width	Depth
Definition	Number of neurons in a hidden layer	Number of hidden layers.
Learning	Learns more features	Learns complex hierarchical features.
Complexity	Higher memory usage	Longer training time
Performance	Better capacity, may overfit	Better accuracy for complex tasks.

Conclusion:

Representation learning enables neural networks to automatically learn useful features from data.

3A) Introduction

Activation function add non-linearity to neural networks, enabling them to learn complex patterns. ReLU, Leaky ReLU (LReLU), and ELU are commonly used activation functions. They differ in how they handle negative input values.

Comparison table:

Feature	ReLU	Leaky ReLU	ELU
Behavior	output 0 for negative values	Gives a small negative output	Gives smooth negative output.
Advantages	Fast and simple	Prevents dying ReLU	Faster convergence better accuracy.
Disadvantages	Dying ReLU problem	Needs α value	More computationally expensive.
Use cases	CNNs, image classification	Deep neural networks	Image, speech, and NLP applications.

Mathematical Behavior:

- ReLU: $f(x) = \max(0, x)$
- Leaky ReLU: $f(x) = x$ if $x > 0$, otherwise αx
- ELU: $f(x) = x$ if $x > 0$, otherwise $a \cdot (e^x - 1)$

Conclusion:

- ReLU is simple and fast.
- Leaky ReLU avoids the dying ReLU problem.
- ELU provides smoother learning and better performance but requires more computation.

HA Introduction:

Unsupervised learning is a type of machine learning in which a neural network learns from unlabeled data. It identifies hidden patterns and relationships without knowing the correct output. It is mainly used for clustering, feature extraction, and dimensionality reduction.

Restricted Boltzmann Machine (RBM)

Def: An RBM is a neural network with one visible layer and one hidden layer that learns the probability distribution of input data.

Working Principle:

- I/P is given to the visible layer.
- Hidden layer extracts important features.

Applications:

- Recommendation system
- Feature extraction.

Auto encoder:

An Autoencoder is a neural network that learns to compress and reconstruct input data.

Working principle:

- The Encoder compresses the input.
- The Decoder reconstructs the original input.
- The network minimizes reconstruction error.

Application:

- Image denoising
- Anomaly detection.

Comparison:

Aspect	RBM	Autoencoder
Objective	Learn data patterns	Reconstruct input data
Architecture	Visible + Hidden layer	Encoder + Decoder
Application	Recommendation, feature extraction.	Denoising, data compression.

Conclusion:

Unsupervised learning helps neural networks discover hidden patterns from unlabeled data.

RBMs are mainly used for feature learning and recommendation systems while Autoencoders are widely used for data compression.